

ClauseChain — Technical Memo

Measured Legal Evidence for Digital Trade | UN Global Hackathon: AI for Digital Trade Regulatory Analysis | RDTII Pillars 6, 7, 8, 9

ClauseChain is an open-source AI pipeline that maps digital trade regulation to the UN RDTII framework. Its thesis is that citation alone is not enough: a real quote can come from a repealed act, a non-binding guideline, an unofficial translation, or a clause whose exception changes the rule. ClauseChain verifies the whole evidence chain — span, source authority, current-law status, legal structure, predicate meaning, and counter-evidence — before any output ships.

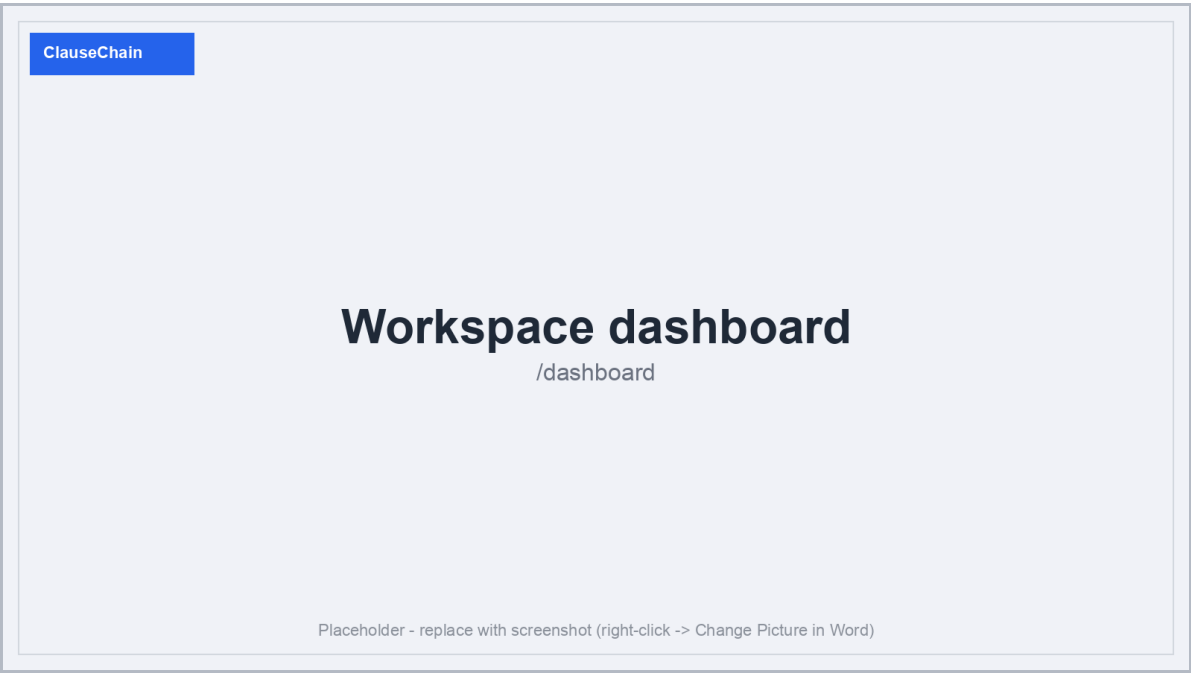


Figure 1 — ClauseChain workspace dashboard.

1. Proposed System Architecture

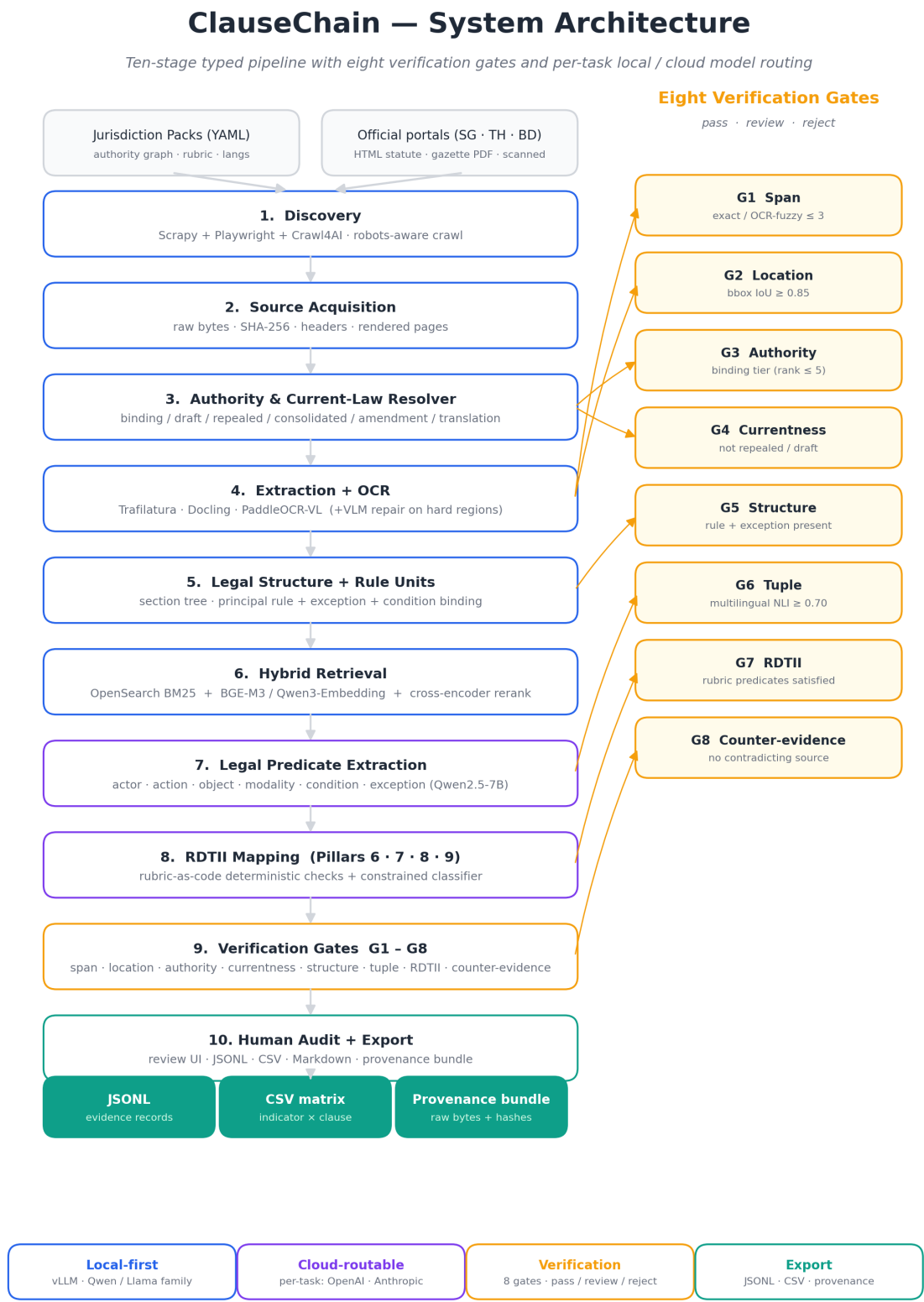


Figure 0 — ClauseChain system architecture: ten-stage typed pipeline (left), eight verification gates G1-G8 mapped to the stage that produces the evidence each gate checks (right), and the local/cloud routing legend.

ClauseChain is a ten-stage typed pipeline (Figure 0). Each stage emits a structured artifact and a measurable quality signal:

Stage	Responsibility
0 Jurisdiction Pack	authority hierarchy, seed sources, citation patterns, language config, RDTII rubric
1 Discovery	polite crawl of official statute databases, gazettes, regulator portals
2 Acquisition	raw bytes, SHA-256, HTTP headers, rendered page images
3 Authority Resolver	binding / draft / repealed / consolidated / amendment / translation classification
4 Extraction + OCR	HTML / native PDF / scanned PDF with text, bbox, per-token confidence
5 Structure + Rule Units	section tree plus principal-rule + exception + condition binding
6 Retrieval	hybrid BM25 + multilingual dense + cross-encoder rerank
7 Predicate Extraction	actor / action / object / modality / condition / exception tuple
8 RDTII Mapping	rubric-as-code deterministic checks + constrained classifier
9 Verification Gates	eight gates G1-G8 (span, location, authority, currentness, structure, tuple, RDTII, counter-evidence)
10 Audit + Export	review UI, JSONL / CSV / Markdown / provenance bundle

The application surface lives at /dashboard plus six pipeline routes: /pipeline/{crawl, harvest, extract, map, trace, export}. A working Next.js 16 / React 19 / TypeScript prototype is in-repo and is what every figure in this memo screenshots.

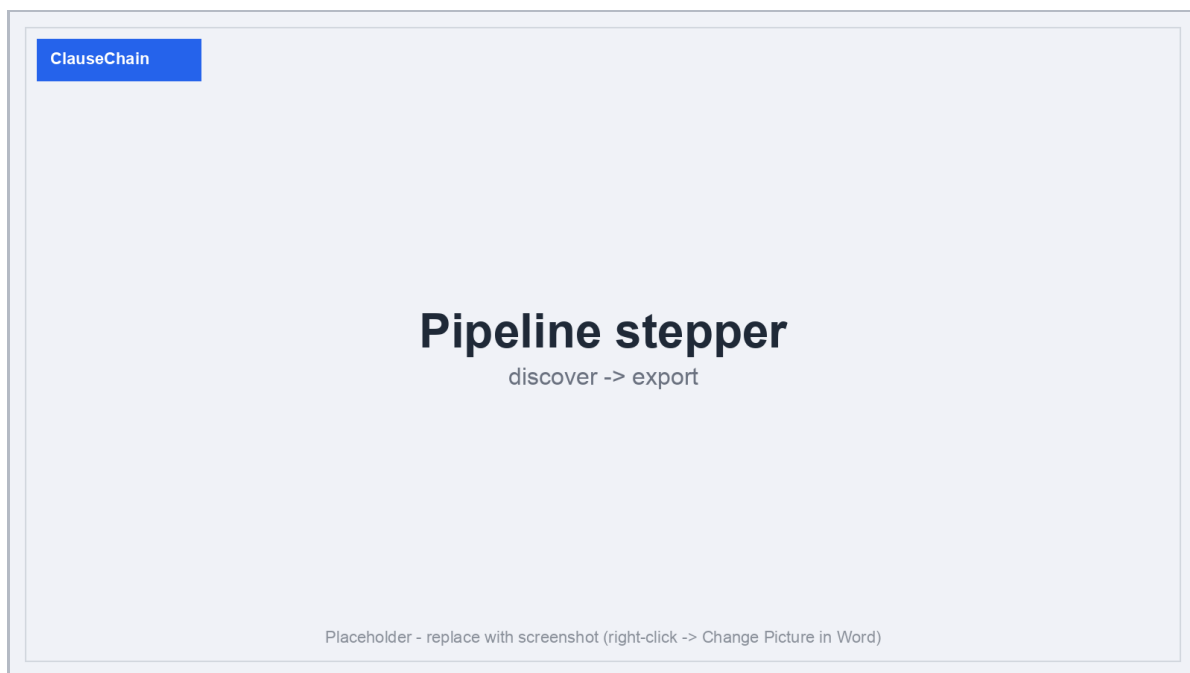


Figure 2 — Nine-stage pipeline stepper (discover -> export) shown across all pipeline pages.

2. Tools, Models, and Methods

ClauseChain is local-first on a single L40S-class GPU, with per-task optional cloud routing to OpenAI or Anthropic. All demo documents are public legal text from official portals, so operators may safely mix local and cloud per pipeline step.

Task	Local default	Cloud option
Crawling	Scrapy + Playwright + Crawl4AI	n/a (deterministic)
HTML extraction	Trafilatura + legal DOM parser	n/a
PDF extraction	Docling + PyMuPDF	n/a
OCR + bbox	PaddleOCR-VL / PaddleOCR	n/a (coordinate-native)
OCR repair (VLM)	Qwen2-VL-7B	gpt-4o or claude-sonnet-4-6
Embeddings	BGE-M3 / Qwen3-Embedding	text-embedding-3-large
Reranking	BGE-reranker-v2-m3	LLM-as-judge
Predicate extraction	Qwen2.5-7B / Llama-3.1-8B-Instruct	gpt-4.1 or claude-sonnet-4-6
RDTII mapping	Qwen2.5-7B	gpt-4.1 or claude-sonnet-4-6
NLI verifier (G6/G7)	DeBERTa-v3 multilingual NLI	gpt-4o or claude-sonnet-4-6

Serving stack: vLLM for local LLMs; FastAPI services; Celery / RQ workers; Postgres + pgvector; OpenSearch (BM25) + Qdrant (dense); MinIO object storage; Next.js audit UI; Docker Compose for the prototype. Routing modes per task: `always_local`, `always_cloud`, `confidence_threshold`, `confidence_margin`.

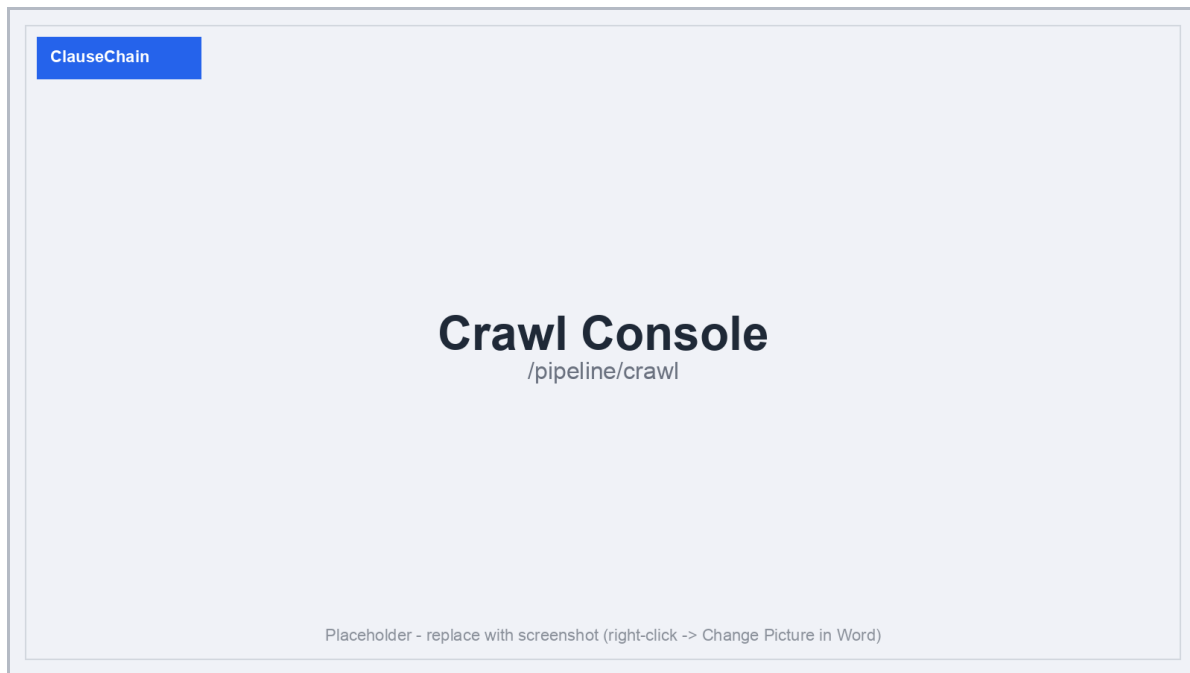


Figure 3 — Crawl Console: politely-crawled candidate documents with type tags (HTML statute, native PDF, scanned PDF, regulator guideline, blocked).

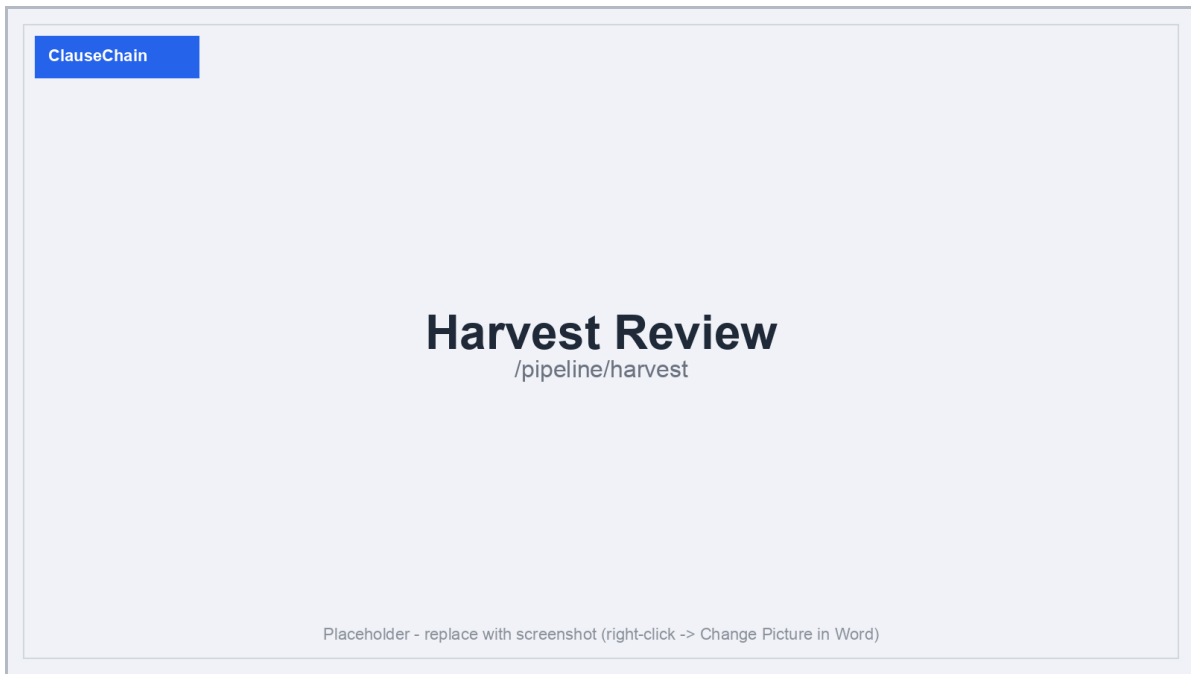


Figure 4 — Harvest Review: human keep/discard before expensive extraction begins.

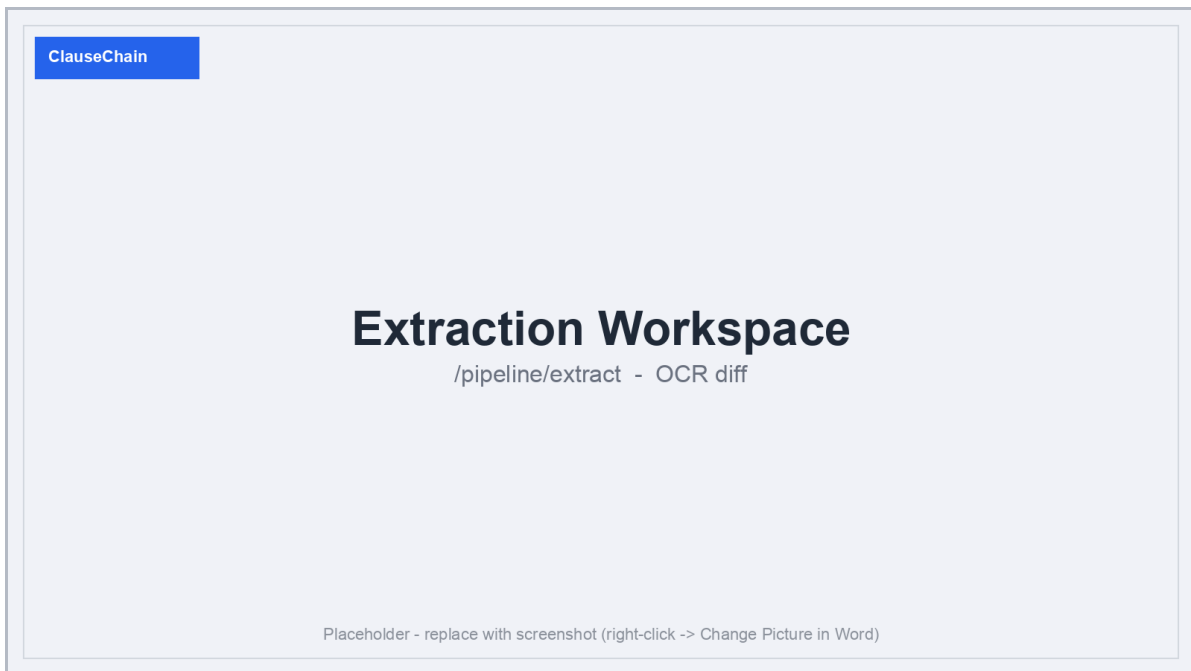


Figure 5 — Extraction Workspace OCR diff: PaddleOCR-VL vs. Tesseract disagreement on a scanned Bengali page, with edit distance and confidence.

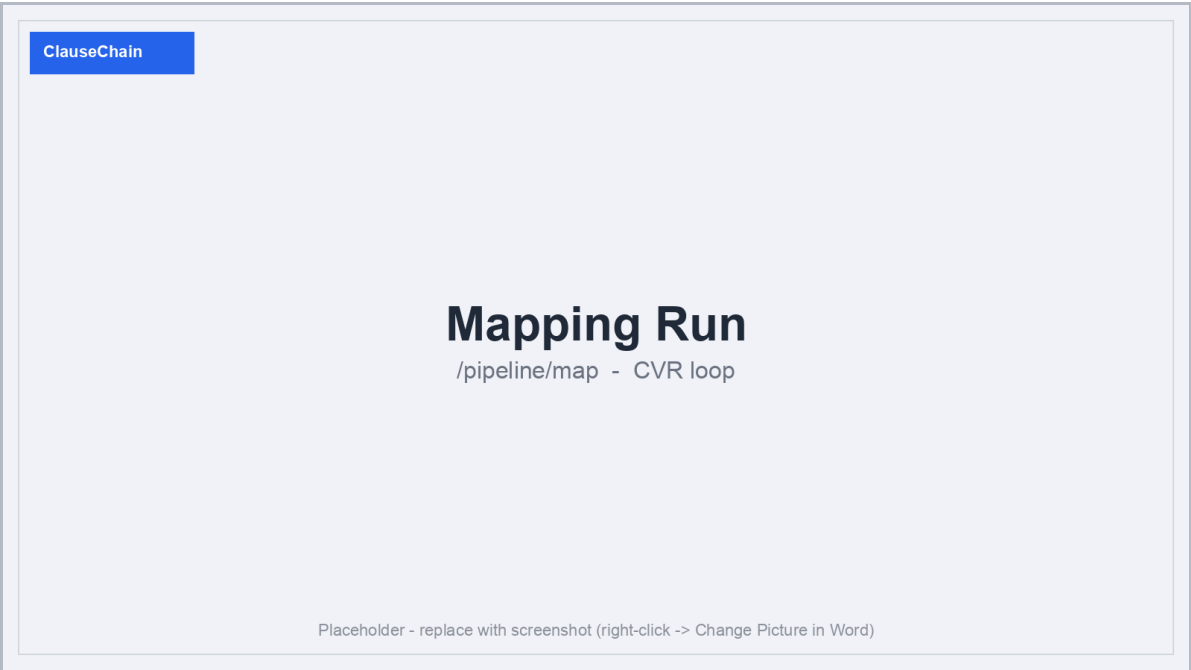


Figure 6 — Mapping Run: L0-L3 autonomy selector and the CVR (Cite / Verify / Reject) loop streaming clauses in real time.

3. Accuracy, Transparency, and Cost-Efficiency

Accuracy — eight verification gates

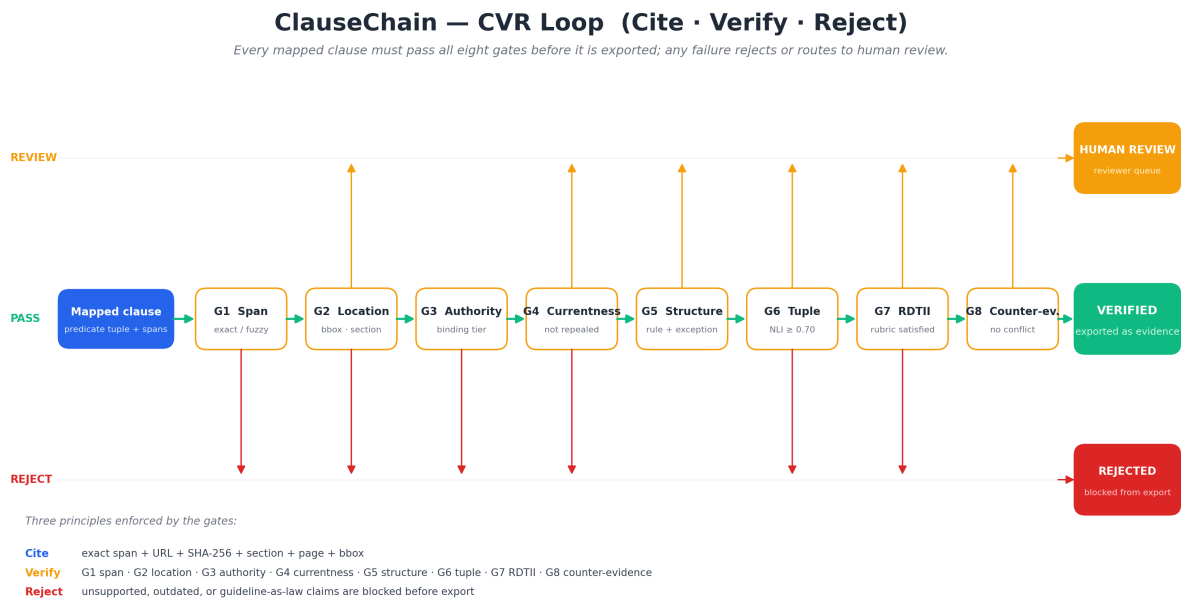


Figure 9 — CVR loop. A mapped clause flows left-to-right through the eight gates on the PASS lane; any uncertain outcome routes up to the REVIEW lane, any disqualifying result drops to the REJECT lane. Only PASS through all eight gates produces a VERIFIED exportable record.

Figure 9 shows the CVR loop. Every claim runs G1 Span (exact byte match, or OCR-fuzzy edit distance ≤ 3 in OCR regions only), G2 Location (bbox IoU ≥ 0.85 or section-path match), G3 Authority (binding tier rank $\leq$

5; guidelines never support binding claims), G4 Currentness (not repealed, draft, or superseded), G5 Structure (correct legal unit with exception present), G6 Tuple (multilingual NLI ≥ 0.70 across populated predicate fields), G7 RDTII (required rubric predicates present, no exclusion triggered), G8 Counter-evidence (corpus search for repeal, amendment, conflict, definition shift). Any failure rejects or routes to human review. Per-stage metrics — discovery recall@20, OCR CER, section-boundary F1, retrieval recall@20, tuple field accuracy, pillar macro-F1, citation exact-match, abstention calibration — are reported with bootstrap confidence intervals against five baselines.

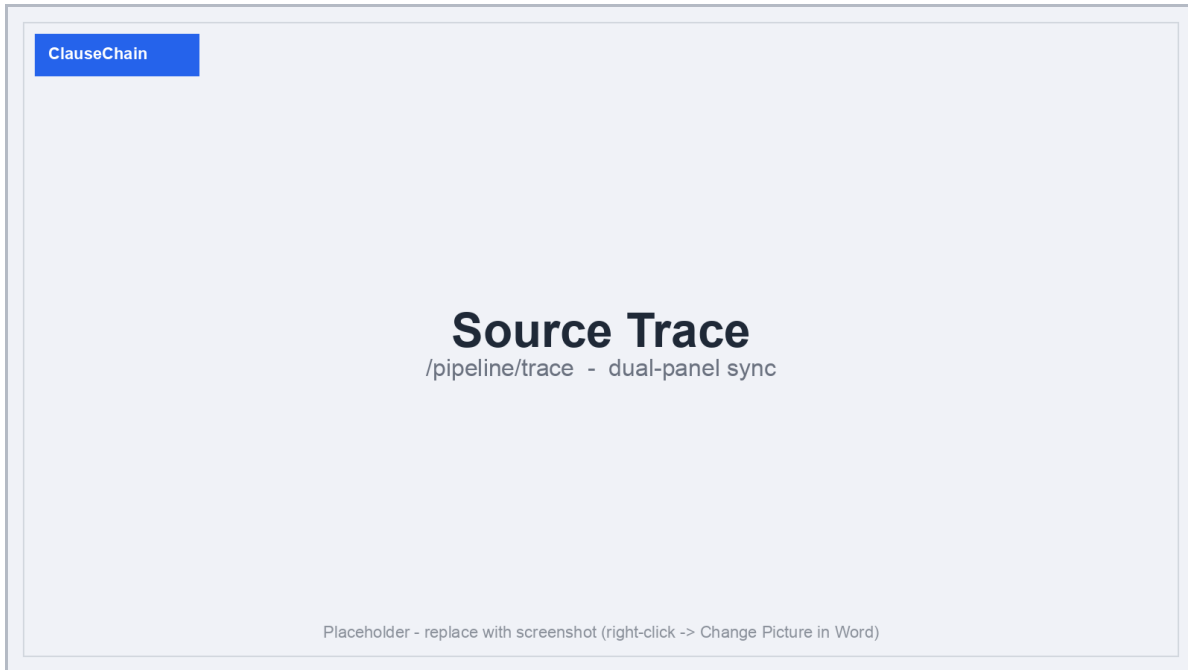


Figure 7 — Source Trace: dual-panel synchronized scroll between extracted clauses and the original source, with per-gate dots, confidence, and match type.

Transparency — provenance bundle

Every exported record carries: source URL, raw-source SHA-256, retrieval timestamp, section, page, char offsets, bbox where available, verbatim quote, span SHA-256, OCR confidence, the eight gate results, and model identity (e.g. local:Qwen2.5-7B or cloud:anthropic:claude-sonnet-4-6). A second analyst on a different machine can re-verify every claim from the provenance bundle alone.



Figure 8 — Export Output: machine-readable records (JSON / CSV / JSONL) with discovery tags, CVR gate results, and verbatim citations.

Cost-efficiency

Local-first default avoids per-token cost entirely. Expensive VLM repair runs only on low-confidence OCR regions; cross-encoder reranking and NLI verification run on small candidate sets, not the full corpus. Cloud escalation is opt-in, per-task, and capped by a per-run USD budget; when the cap hits, the pipeline falls back to local automatically and logs the cap event. Documents tagged 'restricted' in the jurisdiction pack refuse cloud routing entirely. Per-run cost is reported by task and provider so operators can tune the local-vs-cloud split per pipeline.